

Jingyang Zhang

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EDUCATION

Duke University, Durham, NC, USA

Aug 2019 – Dec 2024

- Ph.D. in Electrical and Computer Engineering
 - Advisor: Prof. Yiran Chen, Prof. Hai (Helen) Li

Tsinghua University, Beijing, China

Sep 2015 – Jul 2019

- B.S. in Electronic Engineering

PROFESSIONAL EXPERIENCE

Research Scientist - Post-Train, AI Safety @ Meta Superintelligence Labs

Jul 2026 – Current

- Developed **RL training tasks and data for cyber post-training**, spanning defensive cyber capabilities and appropriate refusal of harmful cyber requests.
 - Constructed agentic cybersecurity tasks for **vulnerability discovery, proof-of-concept generation, and patching**; validated end-to-end **rollout and RL training pipelines** and filtered tasks by difficulty using multi-rollout success rates.
- Developed and deployed **LLM-based cyber safety guardrails** for Meta's **model APIs and Muse** to detect unsafe cyber misuse.
 - Designed a **latency-efficient structured prediction format** that emits the **binary safety decision in the first generated token**, followed by risk-category classification and reasoning, enabling production inference to terminate after the first token when only a binary decision is required.
 - Fine-tuned guard models with **SFT** using public datasets, an online data flywheel, and **adversarial red-teaming data**.
- Conducted **safety and preparedness evaluations for Muse**, assessing cyber risks of personalized agents through **adversarial red-teaming**.
 - Evaluated both **harmful cyber assistance** and **Muse-specific security threats**, including sandbox escape and credential exfiltration attempts against the agent's isolated runtime environment; contributed experiments and analysis to Muse preparedness assessments.

Research Scientist - AI Safety @ Virtue AI

Aug 2025 – Jul 2026

- Co-owned **end-to-end SFT pipeline** for a unified guardrail (prompt and agent action), including data curation, training, evaluation, and efficiency improvements.
- Designed and deployed **agent emulation and adversarial testing frameworks** which is used to conduct **LLM safety evaluation and red-teaming** for real-world enterprise agents, delivering actionable failure analysis across large-scale deployments (Fortune 500-level customers).

Machine Learning Engineer - Foundation Models, Pre-Train @ Sciforium

Jan 2025 – Aug 2025

- Led design of a **purpose-built native multimodal LLM** treating **text, image, video, and audio as first-class modalities** — unified byte-level representation with no modality-specific encoders, targeting coherent cross-modal understanding and generation.
- Designed **model architectures and efficient components** (non-complex RoPE, optimized KV cache, FP8 GEMM) enabling scalable training and inference across all modalities.
- Built **end-to-end pretraining pipelines** with scalable data transformation from raw multimodal inputs (text, image, video, audio) to unified byte representations.
- Conducted **distributed pretraining across 64 GPUs on 8 nodes** using JAX.

PUBLICATIONS

SELECTED CONFERENCE AND JOURNAL PAPERS

- Which Agent Causes Task Failures and When? On Automated Failure Attribution of LLM Multi-Agent Systems
 - *ICML'25 Spotlight* | [paper] [code]
- Unsolvable Problem Detection: Evaluating Trustworthiness of Vision Language Models
 - *ACL'25, ICLR'24 R2FM Workshop* | [paper] [code]
- Min-K%++: Improved Baseline for Detecting Pre-Training Data from Large Language Models
 - *ICLR'25 Spotlight* | [paper] [code] [project page]
- OpenOOD v1.5: Enhanced Benchmark for Out-of-Distribution Detection
 - *Journal of Data-Centric Machine Learning Research, NeurIPS'23 DistShift Workshop Oral* | [paper] [code]

- Mixture Outlier Exposure: Towards Out-of-Distribution Detection in Fine-Grained Environments
 - WACV'23 | [\[paper\]](#) [\[code\]](#)
- Privacy Leakage of Adversarial Training Models in Federated Learning Systems
 - CVPR'22 *The Art of Robustness Workshop Oral* | [\[paper\]](#) [\[code\]](#)
- DVERGE: Diversifying Vulnerabilities for Enhanced Robust Generation of Ensembles
 - NeurIPS'20 Oral | [\[paper\]](#) [\[code\]](#)

OPEN-SOURCE SOFTWARE

- 🔗 **OpenClaw-RL:** Training agents simply by talking through OpenClaw setup (**5k+ Stars**)
- 🔗 **Imms-finetune:** Lightweight codebase for fine-tuning various multimodal (vision) LLMs (**368 Stars**)
- 🔗 **VLM-Visualizer:** Visualizing the attention of vision LLMs (LLaVA) (**284 Stars**)
- 🔗 **OpenOOD:** Large-scale, unified evaluation platform for out-of-distribution detection (**1k+ Stars**)

TECH STACK

Deep Learning Framework and Library: PyTorch, JAX/Flax, transformers, diffusers
Programming Language and Tool: Python, Bash, Bazel (Blaze), Docker, Git